

CONFIDENTIAL · SENTRA STAFF ONLY

Sentra Suite Internal Technical Reference.

The single source of truth for what we sell, what's inside it, how we deploy it, and what it costs. Read by every engineer, salesperson, and director before their first client conversation.

DOC OWNER Engineering · Sentra HQ	LAST UPDATED May 2026
AUDIENCE Staff · Directors · New hires	CLASSIFICATION CONFIDENTIAL

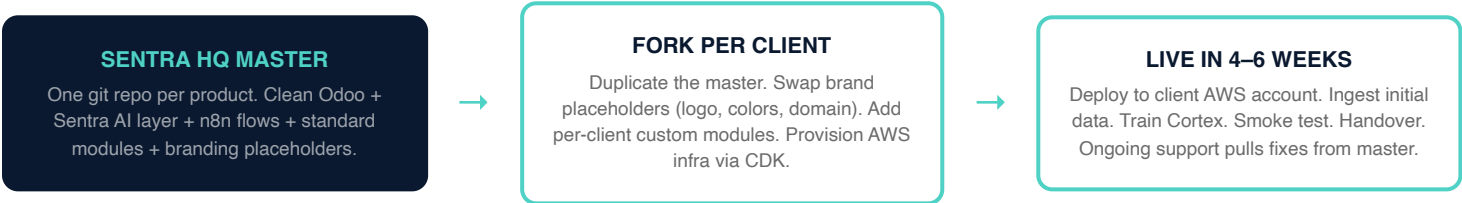
Company thesis

Sentra is an Enterprise Software Builder. We do not write enterprise software from scratch. We don't sell SaaS subscriptions. We take Odoo Community as the universal business platform, layer Sentra's AI + automation stack on top, brand it as the customer's own system, and deploy it on AWS — turning what used to be a 6–12 month custom build into a 4–6 week branded delivery.

The four flagship products

BRAND	WHAT IT IS	DEFAULT OSS BASE
Atlas	Full business operating system — sales, inventory, accounting, HR, projects, fleet.	Odoo Community 17 (full module suite)
Echo	Customer support & omnichannel helpdesk with WhatsApp + AI bot.	Chatwoot (default) + Zammad (IT desk variant)
Cortex	Private AI Brain — LLM trained on customer data, callable from anywhere.	Ollama + OpenWebUI (white-label) + Outline + Qdrant + LangChain
Lens	Executive analytics, KPI dashboards, AI alerts, WhatsApp briefings.	Metabase (default) · Superset (analyst variant)

The master-duplicate model (operational core)



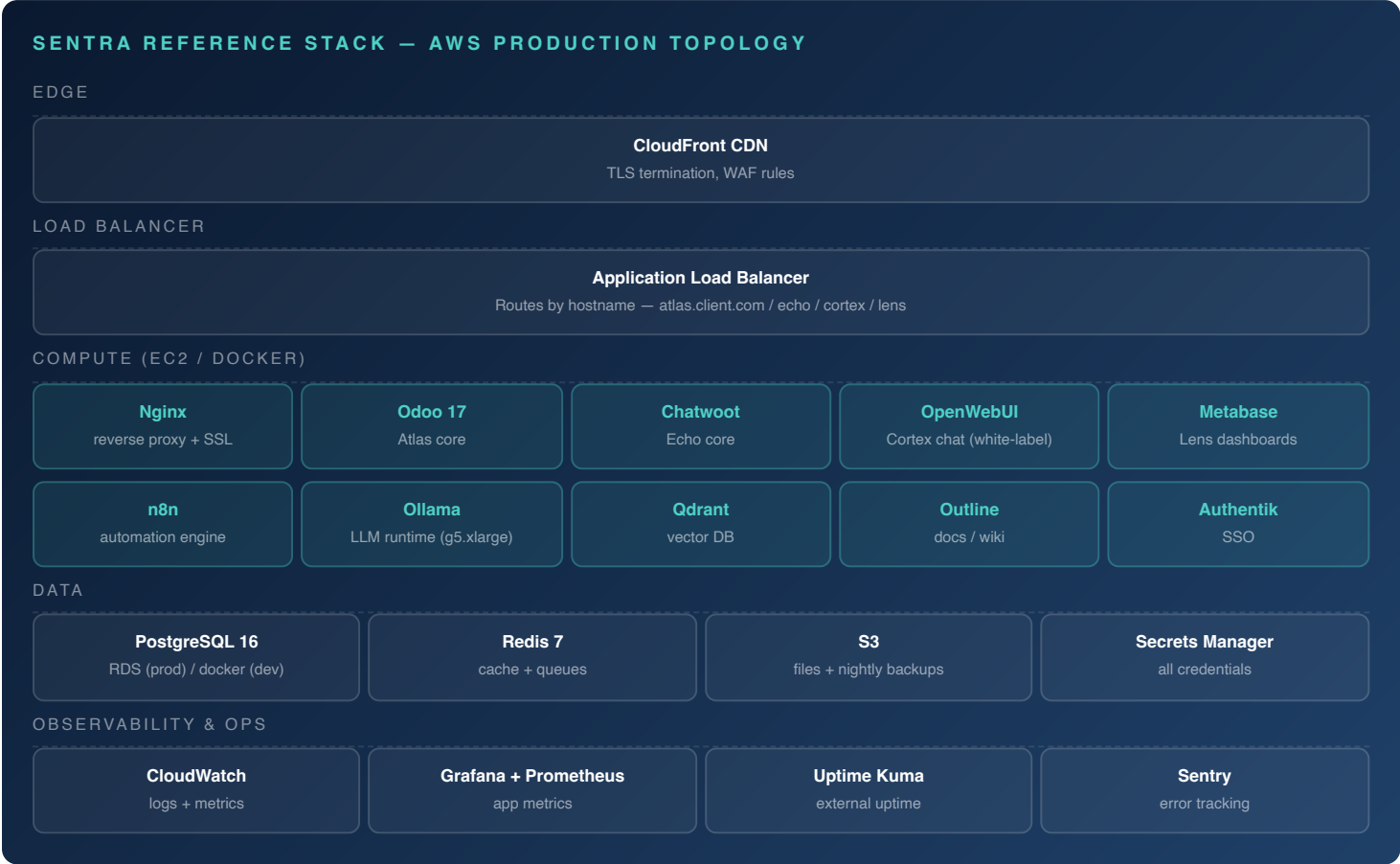
Sales motion in one paragraph (memorise this)

PITCH "We don't write your software from scratch — that's why we're 10x faster and 5x cheaper than a custom build. We take the most battle-tested business platform in the world (Odoo, used by 12 million people) and layer our private AI brain on top. Then we make it look and feel like software your team built itself. Live in weeks. You own the IP. No lock-in." **(Never say this verbatim to a client; it's the internal frame. Client-facing wording is in the Client Brochure.)**

Where we sell · ICP

- **Sweet spot:** 50–500 employee operations companies in services, trading, manufacturing, healthcare, logistics, professional services. USA / EU / UK / Middle East / South Asia.
- **Avoid (red flags):** <20 employees (too price sensitive), heavily regulated finance/insurance unless SOC 2 budget is real, anyone demanding on-prem-only without sponsoring the infra cost.
- **Strongest fit:** companies actively bleeding from one of the five pains in the Client Brochure — especially "tool sprawl" and "drowning customer service".

Every Sentra deployment is a variation on the same reference architecture below. Every engineer should be able to reproduce this from memory before they're cleared to lead a deploy.



Standard ports (internal)

SERVICE	PORT	NOTES
Nginx (HTTPS)	443	Public. Issues Let's Encrypt cert via Certbot.
Odoo (Atlas)	8069	Bound to localhost. Reverse-proxied by Nginx.
Chatwoot (Echo)	3000	Bound to localhost. Reverse-proxied by Nginx.
OpenWebUI (Cortex)	8080	White-labeled build. Localhost only.
Ollama API	11434	Localhost only. Never exposed.
Qdrant	6333	Localhost only. Auth via API key from Secrets Manager.
Outline	3001	Localhost only. Reverse-proxied at <code>/wiki</code> .
Metabase (Lens)	3030	Localhost only. Reverse-proxied at <code>/lens</code> .
n8n	5678	Localhost only. Reverse-proxied at <code>/n8n</code> .
PostgreSQL	5432	RDS in prod (private subnet). Docker in dev.
Redis	6379	Localhost only. Auth via password from Secrets Manager.

RULE Nothing except Nginx / ALB is ever exposed to the public internet. All inter-service traffic is on the localhost loopback or the VPC private subnet. **If you find yourself opening a security group to 0.0.0.0/0 for any service other than 80/443, stop and ask the lead engineer.**

Environment matrix

ENVIRONMENT	WHERE	PURPOSE	WHO PROVISIONS
Local dev	Engineer laptop · Docker Compose	All product development & per-client customisation work happens here first.	Engineer
Sentra staging	Sentra AWS account · eu-west-1	Internal QA, demos, integration tests against master branches.	DevOps lead
Client production	Client AWS account (preferred) or Sentra-managed	Live client deployments. Always isolated per client.	DevOps + project lead

EC2 sizing tiers (per product, prod)

TIER	ATLAS / ECHO / LENS	CORTEx (GPU)	USERS	MONTHLY COST TARGET
Small	t3.medium (2vCPU/4GB)	g5.xlarge (1xA10G)	<25	\$120 + \$480 (Cortex)
Medium	m5.large (2vCPU/8GB)	g5.2xlarge (1xA10G/32GB)	25–100	\$220 + \$920 (Cortex)
Enterprise	m5.xlarge or c5.2xlarge	g5.4xlarge or g6.2xlarge	100–500	\$420+ · \$1,800+ (Cortex)

Costs above are reserved-instance estimates. We pass all infra cost through to the client at AWS list + 10% management fee, billed monthly via SEPA / wire / ACH.

VPC layout (every client deploy)

- 1 VPC per client, CIDR `10.X.0.0/16` where X is allocated from the IPAM doc.
- Public subnets: ALB only.
- Private subnets (compute): all EC2 instances, NAT-gateway egress only.
- Private subnets (data): RDS, ElastiCache. No internet route.
- Security groups: ALB-SG (80/443 from world) → APP-SG (80/443 from ALB-SG only) → DATA-SG (5432/6379 from APP-SG only).

RDS vs self-hosted Postgres — decision matrix

USE RDS WHEN	USE SELF-HOSTED PG (DOCKER ON EC2) WHEN
Client production. Any PII / financial data. >10 users. Compliance mentioned (HIPAA/SOC2/GDPR).	Local dev. Sentra staging. Single-engineer prototypes. Demos that get torn down within 30 days.

Backup & recovery defaults

ASSET	BACKUP METHOD	FREQUENCY	RETENTION	RPO / RTO
RDS Postgres	Automated snapshot	Daily 02:00 UTC	30 days	RPO 24h / RTO 1h
S3 file storage	Versioning + cross-region replication	Continuous	90 days	RPO <5min / RTO 15min
EC2 root volumes	EBS snapshot via Data Lifecycle Manager	Daily	7 days	RPO 24h / RTO 30min
Qdrant vector store	Snapshot to S3 via cron + n8n verifier	Daily 03:00 UTC	14 days	RPO 24h / RTO 2h (re-ingest)

Standard naming conventions

- S3 buckets: `sentra-{client-slug}-{env}-{purpose}` e.g. `sentra-acme-co-prod-backups`
- Hostnames: `atlas.{client-domain}`, `echo.{client-domain}`, `brain.{client-domain}` (Cortex), `insights.{client-domain}` (Lens)
- Git repos: `sentra/client-{slug}-{product}` (private; forked from `sentra/master-{product}`)
- Secrets: AWS Secrets Manager path `/sentra/{client-slug}/{product}/{secret-name}`

Atlas is the Sentra business operating system — the full Odoo Community 17 module suite, plus Sentra custom modules, plus standard n8n flows, plus the WhatsApp + AI hooks shared with Echo and Cortex. This is our highest-revenue product and the most common starting point for a client engagement.

Default Odoo Community 17 modules deployed

MODULE	DEFAULT ON/OFF	NOTES
CRM	ON	Always. Linked to Echo for inbound conversation → lead conversion.
Sales	ON	Quotes, orders, sales teams.
Invoicing & Accounting	ON	Use Community Accounting unless client needs Studio → quote upgrade.
Inventory / Purchase	PER CLIENT	ON for trading/manufacturing; OFF for pure services.
Project	ON	Tasks, timesheets, milestones.
HR / Employees / Time-off	ON	Includes Recruitment for clients hiring 5+ /yr.
Fleet	PER CLIENT	ON for logistics / delivery / field-services clients.
Manufacturing (MRP)	PER CLIENT	ON for actual manufacturers only. Adds complexity — quote uplift.
E-commerce + Website	PER CLIENT	ON for D2C/B2B web sales clients.
Marketing (Email/SMS)	ON	Always. Email goes through Amazon SES.
Helpdesk (base)	OPT-OUT	Disable when shipping with Echo (Chatwoot replaces it).

Sentra custom modules (always installed)

- **sentra_brand_kit** — replaces Odoo's logo, color theme, login screen, email templates, PDF report header/footer with the client's branding via a single config record.
- **sentra_n8n_bridge** — webhook endpoints + Odoo automated actions to trigger n8n workflows from any model event (create/write/unlink). Used for everything from auto-invoicing to WhatsApp notifications.
- **sentra_whatsapp_messaging** — sends/receives WhatsApp messages from any res.partner record. Talks to the WhatsApp Business API via n8n.
- **sentra_cortex_assistant** — sidebar widget across Odoo: ask the Cortex brain about any record. Uses the customer's private LLM (no data leaves the VPC).
- **sentra_lens_embed** — iframes the relevant Metabase dashboard inside any Odoo screen (Sales pipeline page → sales dashboard, etc.).

Default n8n flows shipped with Atlas

- **invoice-on-payment** — payment received in Stripe / bank webhook → auto-confirm Odoo invoice + send WhatsApp receipt.
- **so-to-whatsapp** — new sales order → WhatsApp confirmation to customer with PDF + tracking link.
- **low-stock-alert** — product below reorder point → WhatsApp + email to purchasing manager with one-click PO.
- **weekly-board-report** — every Monday 07:00 → Cortex generates a 1-page weekly summary → sent to leadership WhatsApp.

Pricing — Atlas standalone

TIER	SETUP (ONE-TIME)	HOSTING + SUPPORT / MONTH	TYPICAL CLIENT
Small (<25 users)	\$9,000 – \$15,000	\$400 – \$700	Trading SME, single-location services firm
Medium (25–100)	\$18,000 – \$32,000	\$800 – \$1,400	Multi-team services co, growing trader
Enterprise (100–500)	\$45,000 – \$90,000	\$1,500 – \$2,500	Multi-entity / multi-location enterprise

Standard sequence for deploying Atlas to a new client AWS account. Estimated time end-to-end (excluding data migration): **2 working days** for Small/Medium tier.

Step 1 — Fork the master

```
# Clone Sentra master, push as new client repo
gh repo clone sentra/master-atlas
gh repo create sentra/client-{slug}-atlas --private --push --source=master-atlas
cd client-{slug}-atlas
cp .env.example .env # fill in client-specific values
```

Step 2 — Provision AWS infra (CDK)

1. Switch CDK profile to client account: `aws sso login --profile client-{slug}`
2. Edit `cdk/config.ts` — client slug, region, tier (small/medium/enterprise), domain.
3. `cdk deploy SentraAtlasStack-{slug}` — provisions VPC, subnets, RDS, EC2, ALB, S3, security groups, Route53.
4. Verify outputs: ALB DNS, RDS endpoint, S3 bucket names — record in client onboarding doc.

Step 3 — Deploy Docker stack

```
ssh ec2-user@{atlas-ec2-ip}
git clone git@github.com:sentra/client-{slug}-atlas.git
cd client-{slug}-atlas
sudo docker compose up -d
sudo docker compose logs -f odoo # wait for "Modules loaded"
```

Step 4 — SSL + DNS

1. Point client DNS: `atlas.{client-domain}` → ALB DNS (CNAME).
2. Issue Let's Encrypt cert: `sudo certbot --nginx -d atlas.{client-domain}`
3. Verify HTTPS: `curl -I https://atlas.{client-domain}` returns `200 OK`.

Step 5 — Brand kit + initial config

1. In Odoo as admin: Settings → Sentra Brand Kit → upload logo (SVG + PNG fallback), set primary/accent colors, set company info.
2. Verify: login screen, email templates, invoice PDF, dashboard top-bar all show client branding.
3. Configure SES sender + DKIM in `Settings → Email Servers`. Verify with test send.
4. Configure WhatsApp Business API credentials in `Settings → Sentra WhatsApp`. Send test message to internal number.

Step 6 — Data migration

- **Customers / suppliers:** CSV import via standard Odoo import wizard.
- **Products:** CSV with columns `name, default_code, list_price, qty_available, categ_id`.
- **Open invoices:** manual import or one-shot script in `scripts/import-open-invoices.py`.
- **Historical accounting:** opening balance journal entry only (don't re-import old transactions).

Step 7 — Smoke tests (must pass before handover)

1. Create a test customer + sales order + invoice. Confirm WhatsApp confirmation sent.
2. Mark invoice paid via Stripe test mode. Confirm n8n flow updates Odoo + sends receipt.
3. Reduce a product stock to below reorder point. Confirm low-stock-alert fires.
4. Run `/health` endpoint — all services green.
5. Trigger nightly backup manually (`aws rds create-db-snapshot`). Confirm S3 backup arrives.

Step 8 — Handover checklist

- Admin credentials transferred via 1Password vault share.
- 2x training sessions for client admin + power users (recorded).
- Runbook URL + on-call rotation shared.
- Monthly support contract signed and Stripe subscription active.
- 30-day post-launch check-in scheduled.

NEVER Hard-code credentials in the repo. Always read from AWS Secrets Manager via the `sentra_brand_kit/utils/secrets.py` helper. **One leak ends the company.**

Echo is the Sentra customer support & omnichannel helpdesk product. Default base is **Chatwoot** (best-in-class for WhatsApp-first customer messaging). For pure internal IT helpdesk deployments we substitute **Zammad**. Both share the same Sentra AI + WhatsApp integration layer.

When to use which base

BASE	CHOOSE WHEN	SKIP WHEN
Chatwoot (default)	Customer-facing support. WhatsApp / IG / FB / web chat / email all in one. SMEs, retail, services, e-commerce.	Pure internal IT desk. Heavy ITIL workflow needs (asset CMDB, change management).
Zammad	Internal IT desk. Asset tracking, incident management, ITIL-style workflows. Multi-tier support orgs.	WhatsApp is the primary channel (Zammad's WA support is weaker).

Channel matrix

CHANNEL	GATEWAY	DEFAULT TIER	NOTES
WhatsApp Business	Meta WhatsApp Business API (direct)	All clients	Per-conversation pricing. Templates required for outbound >24h.
WhatsApp (managed)	Twilio WhatsApp	If client wants managed billing	+30% margin, less setup pain.
WhatsApp (budget)	Baileys (open-source)	Demos / very small clients	RISK Unofficial; use only with disclosure.
Email	Amazon SES inbound + outbound	All clients	SPF + DKIM + DMARC required.
Web chat widget	Chatwoot SDK script	All clients	One-line embed on client site.
Instagram DM / Facebook	Meta Messenger Platform	Per-client opt-in	Requires client FB business mgr access.
SMS	Twilio Programmable Messaging	US/UK clients	10DLC registration required (US).
Voice (optional)	Twilio Voice / Vapi	Add-on	For AI voice agent. Quote separately.

AI ticket responder — standard config

- Routing:** incoming message hits Chatwoot → n8n webhook fires → n8n calls Cortex with conversation history + customer record from Atlas → Cortex returns suggested reply + confidence score.
- Confidence >0.85:** auto-reply sent. Logged with "Sent by AI" tag.
- Confidence 0.5–0.85:** draft reply pre-filled in agent inbox. Agent reviews + sends in one click.
- Confidence <0.5:** escalated to human queue. Cortex still suggests likely intent + relevant docs in sidebar.
- Guardrails:** never auto-quote prices, never auto-confirm refunds >\$X, never reply to anything tagged legal or complaint.
- Knowledge source:** always Cortex (the client's private brain). Never general internet search. PII redaction on by default.

Standard SLA tiers shipped

PRIORITY	FIRST RESPONSE	RESOLUTION TARGET	AFTER-HOURS BEHAVIOUR
P1 · URGENT	15 min	4 hrs	WhatsApp on-call escalation
P2 · HIGH	1 hr	1 business day	AI auto-acknowledge, human next morning
P3 · NORMAL	4 hrs	3 business days	AI auto-acknowledge
P4 · LOW	1 business day	1 week	Queued

Pricing — Echo standalone

TIER	SETUP	MONTHLY	WHATSAPP BUSINESS COST
Small (<5 agents)	\$6,000 – \$10,000	\$300 – \$500	Pass-through Meta pricing + 10%
Medium (5–20 agents)	\$12,000 – \$22,000	\$600 – \$1,000	Pass-through Meta pricing + 10%
Enterprise (20+ agents)	\$28,000+	\$1,200+	Custom — usually direct Meta contract

Standard sequence for Echo deployment. Most steps are common with Atlas; only Echo-specific actions are listed. Estimated time end-to-end: **1.5 working days** + WhatsApp Business API approval lag (3–7 days).

Step 1 — Fork & provision (same as Atlas)

```
gh repo create sentra/client-{slug}-echo --private --source=master-echo
cd cdk && cdk deploy SentraEchoStack-{slug}
```

Step 2 — Chatwoot stack up

```
ssh ec2-user@{echo-ec2-ip}
git pull
sudo docker compose up -d # brings up: chatwoot-rails, chatwoot-sidekiq, postgres, redis, n8n
# Run rails migration on first boot
sudo docker compose exec rails bundle exec rails db:chatwoot_prepare
```

Step 3 — WhatsApp Business API setup (Meta direct)

1. Client creates Meta Business Manager account (we cannot do this for them — legal owner).
2. Client adds Sentra as a partner with `manage_whatsapp_settings` permission.
3. Verify business via Meta's verification process (5–7 days — allow buffer in timeline).
4. Apply for WhatsApp Business API access. Use template request boilerplate in `docs/wa-bm-application.md`.
5. Once approved: register phone number (must NOT be already on consumer WhatsApp).
6. Generate permanent System User token. Store in Secrets Manager `/sentra/{slug}/echo/wa-token`.
7. Configure webhook in Meta dev console → `https://{echo-client-domain}/webhooks/whatsapp`.
8. Create at least 3 message templates (welcome, order-confirm, support-handoff). Submit for Meta approval (24–48hrs).

Step 4 — Channel wire-up in Chatwoot

- Settings → Inboxes → New — one inbox per channel (WhatsApp, Email, Web Widget, etc.).
- Email inbox → configure SES inbound rule to forward to `support+{inbox-id}@{client-domain}`.
- Web widget → copy embed snippet to client website team.

Step 5 — AI responder activation

1. In n8n, import flow `echo-ai-responder.json` from `sentra/master-flows`.
2. Set Cortex endpoint URL + API key (from Cortex deploy on adjacent EC2 / VPC peer).
3. Set confidence thresholds in flow settings (defaults: 0.85 / 0.5).
4. Add guardrail rules: refund cap, legal-tag block, language whitelist.
5. Activate flow. Test by sending message from internal WhatsApp number.

Step 6 — Atlas + Cortex integration

- Configure Chatwoot → Atlas customer sync via `sentra_n8n_bridge`: every new contact in Chatwoot creates/links a `res.partner` in Atlas Odoo.
- Configure Cortex — ingest Chatwoot conversation history nightly into Cortex vector store. Helps AI replies stay on-brand.

Step 7 — Smoke tests

1. Send WhatsApp to client number from your phone → conversation appears in Chatwoot.
2. AI responder either auto-replies (if confident) or drafts reply.
3. Reply manually → arrives back on your WhatsApp within 5 seconds.
4. Send email to `support@{client-domain}` → appears as new conversation.
5. Verify Atlas `res.partner` created/linked. Verify dashboard counts increment.

Common issues & fixes

SYMPTOM	LIKELY CAUSE	FIX
WA messages don't arrive	Webhook not registered or signature mismatch	Re-register webhook + verify <code>app_secret</code> in Secrets Manager.
AI replies in wrong language	Cortex prompt not language-aware	Update <code>prompts/echo_responder.md</code> with language detection clause.
SES bounces	SPF/DKIM not propagated	Wait 24h, re-verify with <code>dig</code> .

Cortex is Sentra's private AI brain product. Every client gets a fully isolated LLM stack trained / grounded on their own data. The chat interface is OpenWebUI rebuilt with the client's branding (OpenWebUI's branding is fully stripped — see runbook). Cortex powers the AI assistants in Atlas and Echo, the question-answer chat, and the daily executive summary in Lens.

Component stack

LAYER	COMPONENT	WHY
LLM runtime	Ollama	Self-hosts open weights on EC2 GPU. No data leaves the client VPC.
Default models	Llama 3.1 8B (chat), Mistral 7B (summarisation)	Best quality/cost balance on g5.xlarge.
Heavy-context model	Llama 3.1 70B	Optional. Requires g5.4xlarge or g6.2xlarge minimum.
Specialist models	Mixtral 8x7B, Qwen 2.5 / Qwen 3, DeepSeek, DeepSeek Coder	Loaded on demand. Coder for client-side code generation tasks.
Hybrid (optional)	Anthropic Claude API / OpenAI	For clients with no PII concern + premium quality requirement. Cost is passed through.
Vector DB	Qdrant	Faster + simpler ops than pgvector at scale. Per-client isolated collection.
RAG orchestration	LangChain (Python)	Standard. Custom chains live in <code>sentra/master-cortex/chains/</code> .
Chat UI	OpenWebUI (white-labeled fork)	Mature, theme-able. Sentra fork removes upstream branding.
Docs / wiki base	Outline	Modern UI, markdown-native, easy SSO. Where the human-curated knowledge lives.
Embeddings	nomic-embed-text via Ollama	Local, free, decent quality. Switch to BAAI/bge-m3 for multilingual.
Reranker	BAAI/bge-reranker-v2-m3	Boosts answer relevance ~15%. Optional but recommended.

Document connectors (ingestion sources)

- **Outline pages** — native ingest via Outline API. Re-indexed nightly via n8n.
- **Google Drive** — service account read access, scoped to specific folders.
- **SharePoint / OneDrive** — Microsoft Graph API, app registration in client tenant.
- **Slack** — bot token, ingests selected channels.
- **S3** — bucket of PDFs, docs, slides. Walked daily.
- **Atlas (Odoo) records** — via the standard `sentra_n8n_bridge`. SOPs, customer notes, invoices.
- **Echo conversations** — resolved tickets ingested nightly to improve responder accuracy.

RAG pipeline (default)

1. **Chunking**: 512 tokens with 64-token overlap, semantic-aware splitter for markdown / PDFs.
2. **Embedding**: nomic-embed-text (768-dim) per chunk, stored in Qdrant.
3. **Retrieval**: top-20 by cosine similarity → rerank to top-5 by bge-reranker.
4. **Generation**: top-5 chunks + question + system prompt → Llama 3.1 8B.
5. **Citation**: every answer returns source chunk IDs → rendered as clickable refs in OpenWebUI.
6. **PII redaction**: Microsoft Presidio runs on every chunk before embedding. Configurable allow-list per client.

Inference cost calculator (rule of thumb)

- **Self-hosted on g5.xlarge**: ~\$480/mo fixed regardless of usage (1xA10G GPU). Effective tokens are essentially free at this point.
- **Hybrid Claude/OpenAI usage**: price per million tokens varies; client pays AWS Bedrock or vendor list + 10%.
- **Heavy summarisation jobs (e.g. weekly board report)**: budget ~\$2–\$5/run on hosted models. Negligible on self-hosted.

Pricing — Cortex standalone (add-on to Atlas/Echo also priced like this)

TIER	SETUP	MONTHLY (INCL. GPU EC2)
Small (light usage)	\$6,000 – \$9,000	\$700 – \$1,100
Medium (15–100 daily users)	\$10,000 – \$16,000	\$1,200 – \$1,800
Enterprise (100+, larger model, multi-language)	\$18,000+	\$2,200+

Cortex is the most infrastructure-heavy product because of the GPU EC2. Allow **3 working days** end-to-end (mostly model download + ingestion).

Step 1 — GPU EC2 provisioning

```
cdk deploy SentraCortexStack-{slug}
# spins up: g5.xlarge GPU instance, NVIDIA driver bootstrap, 200GB EBS gp3,
# Qdrant on adjacent t3.medium, security group locked to APP-SG only
```

Step 2 — Ollama install + model pull

```
ssh ec2-user@{cortex-gpu-ip}
curl -fsSL https://ollama.com/install.sh | sh && sudo systemctl enable --now ollama

# Standard set (always)
for m in llama3.1:8b mistral:7b nomic-embed-text; do ollama pull $m; done

# Optional, pull on demand: mixtral:8x7b, qwen2.5:14b, deepseek-coder:6.7b, deepseek-r1:7b
ollama list && curl http://localhost:11434/api/tags # verify
```

Step 3 — Qdrant setup

```
KEY=$(aws secretsmanager get-secret-value --secret-id /sentra/{slug}/cortex/qdrant-key --query SecretString --output text)
sudo docker run -d --name qdrant -p 6333:6333 -v /data/qdrant:/qdrant/storage \
-e QDRANT__SERVICE__API_KEY=$KEY qdrant/qdrant:latest
```

Step 4 — OpenWebUI deploy + branding strip

1. Clone Sentra fork: `git clone git@github.com:sentra/openwebui-whitelabel`
2. Apply client brand kit: `./scripts/apply-brand.sh {slug}` — this script:
 3. Replaces `/static/favicon.png`, `/static/splash.png`, `/static/logo*.svg` with client assets.
 4. Patches `app/manifest.json` name/short_name.
 5. Patches `app/static/i18n/en/translation.json` "Open WebUI" → client product name (e.g. "Atlas Brain" or whatever the client called it).
 6. Patches `tailwind.config.js` color palette to client primary/accent.
 7. Replaces footer "Open WebUI" link → client support URL.
8. Build & run: `docker compose up -d openwebui`.
9. Configure backend in OpenWebUI: Settings → Connections → add Ollama URL `http://{cortex-gpu-ip}:11434`.
10. Visual QA: load login screen, chat screen, settings — **zero "Open WebUI" strings should remain**. Run `grep -r "Open WebUI" /app` to confirm.

Step 5 — Outline (wiki) deploy

```
cd /opt/outline
docker compose up -d
# Configure SSO via Authentik. Apply Sentra brand kit (logo, primary color).
```

Step 6 — Initial ingestion

1. Configure data sources in `cortex/config/sources.yaml` — one block per connector (Outline, Drive, SharePoint, Slack, S3, Atlas).
2. Run initial ingestion: `python -m cortex.ingest --all --client {slug}`. Expect ~30 min–2 hours depending on document volume.
3. Verify Qdrant collection count: `curl -H "api-key: $KEY" http://localhost:6333/collections/{slug}`.
4. Schedule nightly re-index: n8n flow `cortex-reindex`, cron `0 2 * * *`.

Step 7 — Atlas integration (sidebar widget)

- Install `sentra_cortex_assistant` module in Atlas Odoo (already in master).
- Configure Cortex API URL + API key in `Settings → Sentra Cortex`.
- Verify: open any record in Odoo → Cortex sidebar appears → ask "summarise this customer's history" → answer with citations.

Step 8 — Smoke tests & cost monitor

1. Ask 5 questions covering: a customer policy, a SOP, a recent invoice, a Slack thread, a Drive PDF. All must return correct answers with citations.
2. Trigger weekly board report flow manually. Verify Cortex generates 1-page summary → arrives on WhatsApp.
3. Set CloudWatch budget alarm at 110% of expected monthly EC2 cost.

WHITE LABEL The OpenWebUI branding strip is the single most important step. **If a client sees "Open WebUI" anywhere — favicon, browser tab, footer, login screen — the illusion of bespoke software is broken.** Run the brand QA script (`scripts/brand-qa.sh`) before every handover.

Lens is the Sentra executive analytics product. Default base is **Metabase** (best UX for non-technical execs). For clients with analyst-heavy teams who want notebook-style exploration, swap to **Apache Superset**. Lens always pairs with Atlas (it reads from the Atlas Postgres replica). It can also pair with Echo and Cortex telemetry.

When to use which base

BASE	CHOOSE WHEN
Metabase (default)	Execs who want a clean dashboard and the occasional "ask a question" feature. SMEs to mid-market.
Apache Superset	Companies with a data team. Need SQL Lab, Jinja templates, advanced viz, dbt integration.

Standard dashboards shipped (Metabase)

DASHBOARD	AUDIENCE	REFRESH	DEFAULT PANELS
Sales Pulse	Sales leadership	15 min	Pipeline funnel, week-on-week revenue, top-10 deals, conversion rate, lead sources
Operations Live	COO / Ops	5 min	Open orders, WIP, on-time delivery rate, SLA breach count, staff utilisation
Finance Snapshot	CFO	1 hr	Cash position, AR ageing, AP due, MRR, gross margin trend, expense outliers
Customer Health	CS / Account mgmt	15 min	NPS, ticket volume, response time SLA, churn risk score (Cortex), NRR
Inventory Watch	Purchasing	1 hr	Stock-on-hand, reorder triggers, slow movers, supplier lead times
HR & People	HR / CEO	Daily	Headcount, time-off, recruitment funnel, training compliance
CEO Cockpit	CEO	Real time	5 KPIs only — the ones the CEO chose. Plus the AI daily summary.

Data plumbing

- **Source:** Atlas Postgres (Odoos). We create a read replica in RDS so Metabase queries don't impact production.
- **Materialised views:** for any dashboard query taking >3s, create a materialised view refreshed via n8n cron. Lives in schema `sentra_lens`.
- **Echo data:** Chatwoot replica → joined into Customer Health dashboard.
- **Cortex telemetry:** usage stats, popular questions, knowledge gaps → "Brain Insights" board.

AI alert engine

The Lens differentiator vs vanilla Metabase. Implemented entirely in n8n.

- Every scheduled metric query runs through n8n `lens-anomaly-detector` flow.
- Compares current value vs 30-day rolling baseline + 2σ band.
- Anomaly → sent to Cortex with context. Cortex generates a 2-sentence English explanation.
- Delivered via WhatsApp to the metric's owner. Severity tag (info / warning / urgent) in the message.
- Daily 07:00: `lens-daily-briefing` flow — Cortex summarises last 24h across all dashboards into a 60-second exec voice note (Vapi optional) + WhatsApp text.

Embedding into Atlas

- Module `sentra_lens_embed` iframes the relevant Metabase dashboard inside Odoos screens.
- Auth: signed JWT (1-hour expiry) — user logs into Odoos, dashboard appears already authenticated.
- Examples: Sales pipeline screen → Sales Pulse panel; Project page → Project KPIs; Helpdesk page (Echo) → Customer Health.

Pricing — Lens standalone

TIER	SETUP	MONTHLY
Small (5 dashboards, <15 users)	\$5,000 – \$8,000	\$300 – \$500
Medium (10–15 dashboards, alerts on)	\$10,000 – \$15,000	\$600 – \$1,000
Enterprise (multi-source, predictive layer, voice briefing)	\$18,000+	\$1,200+

Lens is the lightest of the four products to deploy. Estimated time: **1 working day**, plus dashboard tuning per client (1–2 days).

Step 1 — Provision read replica

```
# Add a read replica to Atlas RDS
aws rds create-db-instance-read-replica \
  --db-instance-identifier sentra-{slug}-atlas-replica \
  --source-db-instance-identifier sentra-{slug}-atlas-primary \
  --db-instance-class db.t3.medium
```

Step 2 — Metabase up

```
ssh ec2-user@{lens-ec2-ip}
git clone git@github.com:sentra/client-{slug}-lens.git
cd client-{slug}-lens
sudo docker compose up -d # brings up: metabase, postgres (metabase metadata)
# wait for metabase to finish init - check logs for "Metabase started"
```

Step 3 — Connect data source

1. In Metabase admin: Data → Add database → PostgreSQL.
2. Host: replica endpoint. Database: `{slug}_odoo`. User: `lens_readonly` (created by CDK).
3. Verify connection. Sync schema (~5 min).

Step 4 — Import standard dashboards

```
# Sentra dashboards are versioned as Metabase serialised export.
docker exec metabase java -jar /app/metabase.jar load \
  /sentra-dashboards/v1 --on-error continue
# Then re-map each card's database ID to the new connection.
python scripts/remap-database-ids.py --client {slug}
```

Step 5 — Brand kit

1. Apply Sentra Metabase whitelabel patch (Enterprise edition feature, but achievable in Community via custom CSS overlay):
 2. Inject `/app/dist/css/sentra-brand-{slug}.css` via reverse proxy.
 3. Replace logo, favicon, login background.
 4. Hide the "Powered by Metabase" footer via CSS `display: none`.
5. Verify: login screen, dashboard chrome, embedded view all show client branding.

Step 6 — AI alert flows

1. In n8n, import flows `lens-anomaly-detector.json` and `lens-daily-briefing.json`.
2. Configure metric watch list with the client during discovery (which 5–10 numbers does the CEO want pinged about?).
3. Configure WhatsApp recipients per metric (owner mapping).
4. Test by manually injecting an outlier into a test metric → alert should arrive on phone within 60 seconds.

Step 7 — Embed into Atlas

- In Atlas: Settings → Sentra Lens → paste Metabase URL + secret signing key.
- Configure embed mappings (which Odoo screen shows which dashboard panel).
- Verify: open Sales pipeline in Odoo → Sales Pulse panel renders embedded.

Step 8 — CEO cockpit configuration

This is the white-glove step. Sit with the CEO for 30 min, ask "what 5 numbers do you want on your phone every morning?" — then build exactly that dashboard, no more, no less. Set the AI daily briefing to summarise it in 60 seconds.

Smoke tests

1. All 7 standard dashboards render in <3 seconds.
2. One metric anomaly fires correctly to test WhatsApp number.
3. Daily briefing arrives on schedule with correct content.
4. Embedded view in Atlas shows the right dashboard for the right user role.

All workflows live in `sentra/master-flows` and are imported into each client's n8n instance during deploy. If you build a flow more than once, promote it to the master library. Standard flows are versioned (`v1.json`) and changelog'd.

FLOW ID	TRIGGER	NODES	OUTPUT
echo-ai-responder V1	Chatwoot webhook (new message)	Webhook → Atlas customer lookup → Cortex chain (with conv history) → Confidence gate → Auto-send / Draft / Escalate	Reply sent or drafted in Chatwoot inbox
invoice-on-payment V2	Stripe / bank payment webhook	Webhook → Match Atlas invoice → Mark paid → WhatsApp template send	Receipt on customer WhatsApp + Odoo invoice paid
so-to-whatsapp V1	Atlas SO confirmed event	Atlas hook → Render PDF → Upload to S3 → WhatsApp template with link	Customer gets order confirmation + tracking link
low-stock-alert V1	Cron 0 8 * * * (daily 8 AM)	Atlas query: products below reorder → group by supplier → WhatsApp + draft PO link	Purchasing manager WhatsApp summary
lens-anomaly-detector V2	Cron every 15 min	Run watched metric queries → compare to baseline → if outlier → Cortex explain → WhatsApp send	Alert to metric owner with explanation
lens-daily-briefing V1	Cron 0 7 * * * (daily 7 AM)	Pull all dashboard summaries → Cortex compose 60s briefing → WhatsApp send (text + optional Vapi voice)	Exec daily briefing on WhatsApp
weekly-board-report V1	Cron 0 7 * * 1 (Monday 7 AM)	Cortex compose 1-pager → render PDF → email + WhatsApp to leadership	Weekly board PDF in inbox + WhatsApp
cortex-reindex V1	Cron 0 2 * * * (daily 2 AM)	Walk all configured sources → diff against last index → embed new chunks → upsert Qdrant	Cortex stays current
backup-verifier V1	Cron 0 4 * * * (daily 4 AM)	List recent RDS snapshots + S3 backup objects → check freshness + integrity hash → alert if stale	Slack/WhatsApp alert if backup missing or stale
new-lead-router V1	Echo new conversation tagged "sales"	Cortex extract intent + budget signals → create Atlas lead → assign to round-robin → WhatsApp salesperson	Lead in CRM + salesperson notified
sla-breach-watch V1	Cron every 5 min	Echo open conversations → check SLA → if breach imminent → escalate + WhatsApp owner + manager	SLA breaches caught before they happen
onboarding-helper V1-BETA	New employee created in Atlas HR	Cortex generate personalised onboarding plan → create tasks in Project → WhatsApp the new hire welcome	Day-1 onboarding kit auto-deployed

CONTRIBUTING New flows submitted via PR to `sentra/master-flows`. Required: `README.md`, `v{n}.json`, secrets template, two-engineer code review, smoke test in staging. **Do not import un-reviewed flows into client n8n.**

The internal source of truth. **Never share this exact sheet with clients.** Client-facing prices and ranges are in the Client Brochure. Use this to scope, quote, and protect margin.

Setup tiers (one-time, per product)

PRODUCT	SMALL	MEDIUM	ENTERPRISE	NOTES
Atlas	\$9K–\$15K	\$18K–\$32K	\$45K–\$90K+	Add custom modules at \$8K–\$15K each
Echo	\$6K–\$10K	\$12K–\$22K	\$28K+	WhatsApp BSP setup +\$1.5K (one-time)
Cortex	\$6K–\$9K	\$10K–\$16K	\$18K+	Initial corpus >100GB → +\$3K ingest fee
Lens	\$5K–\$8K	\$10K–\$15K	\$18K+	Custom dashboards beyond standard 7 → \$1.2K each
Suite (all 4)	~\$21K	~\$42K	~\$95K+	25% OFF applied automatically

Monthly hosting + support tiers

TIER	ATLAS	ECHO	CORTEX	LENS	SUITE TOTAL
Small	\$400–\$700	\$300–\$500	\$700–\$1.1K	\$300–\$500	~\$1.7K–\$2.8K
Medium	\$800–\$1.4K	\$600–\$1K	\$1.2K–\$1.8K	\$600–\$1K	~\$3.2K–\$5.2K
Enterprise	\$1.5K–\$2.5K	\$1.2K+	\$2.2K+	\$1.2K+	~\$6.1K+

Other line items

ITEM	INTERNAL COST	CLIENT PRICE	MARGIN
Custom Odoo module (1 dev-week)	~\$3.5K loaded cost	\$8K–\$15K	~55%+
Custom n8n flow (medium complexity)	~\$1K loaded cost	\$2.5K–\$5K	~60%
Bespoke Cortex chain / fine-tune	~\$5K loaded cost	\$12K–\$25K	~55%+
AWS infra pass-through	AWS list	AWS list + 10%	10% management fee
Services day rate (senior eng)	~\$700 loaded	\$1.2K–\$1.8K/day	~50%
Training session (per session)	~\$400	\$1.2K	~65%

Payment schedule (default)

- Setup: **50% on kickoff** (signed SOW), **50% on go-live**.
- Monthly hosting + support: invoiced in advance on the 1st, net-15.
- Custom modules / bespoke work: 50/50 against milestone delivery.
- Wire / ACH / SEPA. Stripe for <\$10K invoices. Crypto on request, with 10% surcharge.

First-milestone refund guarantee (always offer)

POLICY "If Milestone 1 doesn't meet the written specification, the milestone payment is waived and the kickoff payment is refunded. You walk away whole."

This is our most important sales-closing tool. **Quoting any deal without it requires partner approval.**

Discount authority matrix

DISCOUNT %	APPROVAL NEEDED
0–10%	Sales lead can authorise
10–20%	Director sign-off
20–25% (suite default)	Auto, no approval needed (suite discount)
>25% off list	Two-partner sign-off, written justification, deal review

Discovery questions (ask all of these on call 1)

- 1. How many people on the team? Across how many locations / entities?
- 2. Which 3 tools / spreadsheets does your team complain about most?
- 3. How do customers reach you today? Which channels? What % is unanswered >24h?
- 4. What's your current monthly software bill (everything — CRM, accounting, support, BI, AI tools)?
- 5. If we built one platform that replaced 80% of those tools and gave you AI on top, what would that be worth annually?
- 6. Who's the executive sponsor? Who'll use it daily? Who's the technical owner on your side?
- 7. Do you have a preferred AWS account or do you want us to host?
- 8. What's the deadline driving this? (Always probe for the real why-now.)
- 9. Have you tried to fix this before? What happened? (Important — previous failures = scars.)
- 10. What does success look like to you 6 months after go-live?

Qualifying yes/no checks

CHECK	PASS	FAIL = WALK
Team size	20–500	<15 (too price-sensitive) or >1000 (too political)
Budget reality	\$15K+ for setup, \$400+/mo	"Just exploring", no number, asks for free PoC
Decision authority	CEO / Founder / COO on the call by week 2	Stuck talking to junior procurement
Timeline	Wants to start within 60 days	"Maybe Q3 next year"
Sponsor pain	Can name 2+ painful symptoms in their own words	Vague "we want AI" or "we want digital transformation"

Red flags (walk politely)

- "We need this in 2 weeks" — no, even our 4-week is the minimum and we don't ship slop.
- "Can we pay you in equity?" — no equity-only deals. Cash + small equity considered for strategic accounts only.
- "We just need a quick PoC for free" — charge for the ASSESS engagement or walk.
- "We'll send you our 70-page RFP" — if they won't take a 30-min call first, they're shopping price not partner.
- "Our IT will install everything" — OK in theory, but watch for them turning into the bottleneck during deploy.
- Refusal to use AWS or refusal to put data on cloud — quote 2x for on-prem and add 90 days to timeline.

Top objections + answers

OBJECTION	ANSWER (SHORT)
"It's too expensive."	Compare to current monthly tool spend × 12 + 1 ops hire × salary. We're cheaper in year 1 and pay back in 5–6 months.
"What if you go bust?"	You own the IP and the AWS account. The platform is yours forever. Hand-off docs included.
"We already have Salesforce / SAP."	Great. We don't replace those. We layer the AI brain (Cortex) and customer comms (Echo) on top, or build the missing pieces. Cherry-pick.
"AI is over-hyped."	Agree it is. Show the boring AI — SLA breach alert, AI-drafted reply approved by human, weekly board summary. Boring + measurable wins.
"We need a PoC first."	Yes — that's what the ASSESS engagement is. \$9K, 3 weeks, fully refunded against TRANSFORM if you proceed.
"How fast really?"	Discovery week 1. Live in week 6. Friday demos every week. You see real progress every 7 days.

Demo script (30-min slot)

- 1. **0–3 min:** Recap their pain in their own words (always).
- 2. **3–5 min:** Show the model in one slide (foundation + AI + brand = your platform). Don't mention Odoo unless asked technical.
- 3. **5–15 min:** Live demo against the relevant pain. Atlas dashboard for tool sprawl; Echo conversation for customer chaos; Cortex Q&A for knowledge silos; Lens dashboard for visibility.
- 4. **15–20 min:** Show the WhatsApp + AI assistant magic moment (their CEO will get a daily briefing on his phone).
- 5. **20–25 min:** Pricing band. Suite discount. Risk-reversal guarantee.
- 6. **25–30 min:** Next step — written proposal in 5 days. Confirm decision-maker on next call.

Glossary & acronyms

TERM	MEANING	TERM	MEANING
Atlas	Sentra's business OS product (Odoo-based)	RPO / RTO	Recovery Point / Time Objective
Echo	Sentra's omnichannel support product (Chatwoot)	SLA	Service Level Agreement
Cortex	Sentra's private AI brain product (Ollama + RAG)	NRR	Net Revenue Retention
Lens	Sentra's exec analytics product (Metabase)	SOW	Statement of Work (signed scope doc)
Suite	All four products sold together (25% off)	ASSESS	Our paid PoC / discovery engagement
Master	The maintained reference repo per product	TRANSFORM	Full build engagement
Brand kit	The client's logo + colors + email templates	PARTNER	Ongoing monthly relationship
RAG	Retrieval-Augmented Generation (Cortex pattern)	White-label	Branded as the client's, no Sentra/upstream marks
BSP	Business Solution Provider (WhatsApp gateway)	ICP	Ideal Customer Profile
SES	AWS Simple Email Service	PII	Personally Identifiable Information

Internal tools we use

TOOL	PURPOSE	WHERE IT LIVES
GitHub Org	All code, masters and client forks	github.com/sentra
1Password	Shared credentials, client vaults	Sentra business account
Notion	Internal docs, meeting notes, client onboarding boards	notion.so/sentra
Linear	Engineering issue tracker	linear.app/sentra
Slack	Internal comms + client shared channels	sentra.slack.com
HubSpot (free)	Sales CRM, deal pipeline, demo scheduling	Sales lead
Stripe	Invoicing for <\$10K transactions	Finance
AWS Org	Multi-account: hq, staging, per-client	DevOps lead

Product owners

PRODUCT	TECH LEAD	SALES LEAD	RUNBOOK LOCATION
Atlas	(assign)	(assign)	Notion » Engineering » Atlas
Echo	(assign)	(assign)	Notion » Engineering » Echo
Cortex	(assign)	(assign)	Notion » Engineering » Cortex
Lens	(assign)	(assign)	Notion » Engineering » Lens
n8n flows	(assign)	n/a	github.com/sentra/master-flows
AWS / DevOps	(assign)	n/a	Notion » DevOps » Runbooks

On-call & incident response

- **Rotation:** weekly, Mon 09:00–Mon 09:00, by SMS + WhatsApp escalation.
- **P1 (client down):** respond <15 min, restore <4 hrs.
- **P2 (degraded):** respond <1 hr, restore <1 business day.
- **Post-mortem:** any P1 within 5 business days, written + shared with affected client + internal team.
- **Status page:** status.sentra.tech — updated for any P1 or scheduled maintenance.

Contact for this document

DOC OWNER Engineering · Sentra HQ · contact.sentra.technologies@gmail.com · WhatsApp +94 70 466 9834. **This document is versioned.**
Suggest changes via PR to [sentra/internal-docs](https://github.com/sentra/internal-docs).